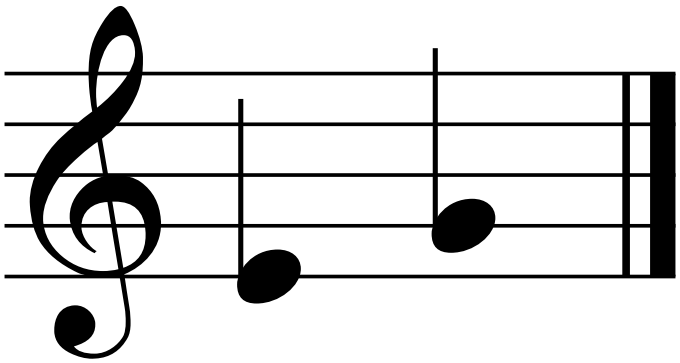
Verovio

Music engraving in Typst



github.com/bernsteining/verovio · typst.app/universe/package/verovio

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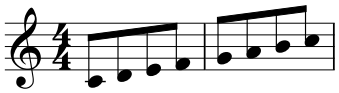
Verovio — Music Engraving Plugin for Typst

Verovio is a music notation engraving library. This Typst plugin wraps Verovio as a WebAssembly module, rendering music from multiple input formats directly into SVG embedded in your document.

Quick Start

Render inline ABC notation:

```typst
#import "@local/verovio:0.1.0": render-music
#render-music("X:1\nM:4/4\nK:C\nCDEF|GABc|")
```

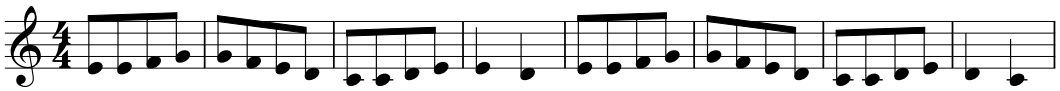


Or define a show rule to render ABC code blocks automatically:

```typst
#import "@local/verovio:0.1.0": render-music
#show raw.where(lang: "abc"): it => render-music(it.text)

```abc
X:1
T:Ode to Joy
M:4/4
K:C
EEFG|GFED|CCDE|E2D2|
EEFG|GFED|CCDE|D2C2|
```
```

Ode to Joy



API Reference

`render-music`

```typst
#render-music(
  data,             // string: music data (ABC, MusicXML, MEI, Humdrum, Volpiano, CMME)
  options: none,    // dictionary: verovio options
  page: 1,          // int: page number to render
  ..args,           // forwarded to Typst's image() (width, height, fit, alt)
)
```

`music-page-count`

```typst
#let n = music-page-count(data, options: none)
```

Returns the number of pages for the given music data and options. Useful to loop over pages of a multi-page score:

```typst
#let data = read("score.musicxml")
#let pages = music-page-count(data)
```

```
#for p in range(1, pages + 1) {
  render-music(data, page: p, width: 100%)
  pagebreak()
}
```

Verovio Options

Options are passed as a Typst dictionary. They map directly to Verovio's toolkit options.

Common		
`adjustPageHeight`	`true`	Crop SVG height to content
`adjustPageWidth`	`false`	Crop SVG width to content
`scale`	`100`	Scale factor (percent)
`font`	`"Leipzig"`	Music font: Leipzig, Bravura, Gootville, Leland, Petaluma
`inputFrom`	`"auto"`	Format: auto, mei, musicxml, abc, humdrum, volpiano, cmme
`pageWidth`	`2100`	Page width (MEI units)
`pageHeight`	`2970`	Page height (MEI units)
`pageMarginTop`	`50`	Top margin
`pageMarginBottom`	`50`	Bottom margin
`pageMarginLeft`	`50`	Left margin
`pageMarginRight`	`50`	Right margin
`landscape`	`false`	Landscape orientation
`breaks`	`"auto"`	Line breaks: auto, line, encoded, none
`condense`	`"auto"`	Condense: auto, none, encoded
`transpose`	`""`	Transpose (e.g. "M2" for major second up)
`header`	`"auto"`	Header: auto, none, encoded
`footer`	`"auto"`	Footer: auto, none, encoded
Layout		
`spacingStaff`	`12`	Spacing between staves
`spacingSystem`	`12`	Spacing between systems
`spacingLinear`	`0.25`	Linear spacing factor
`spacingNonLinear`	`0.6`	Non-linear spacing factor
`unit`	`9`	Base unit size (half staff space)
`stemWidth`	`0.2`	Stem width
`barLineWidth`	`0.3`	Bar line width
`staffLineWidth`	`0.15`	Staff line width
`lyricSize`	`4.5`	Lyrics font size
`hairpinSize`	`3.0`	Hairpin height
SVG Output		
`svgViewBox`	`false`	Use viewBox instead of width/height
`svgRemoveXlink`	`false`	Use href instead of xlink:href
`svgBoundingBoxes`	`false`	Add bounding box rects (debug)
`removeIds`	`false`	Strip element IDs from SVG
`smuflTextFont`	`"embedded"`	SMuFL text font: embedded, linked, none

Full reference: https://book.verovio.org/toolkit-reference/toolkit-options.html

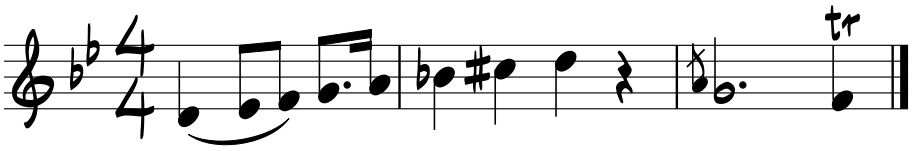
Music Fonts

Five SMuFL-compliant music fonts are available. Set the font with the `font` option:

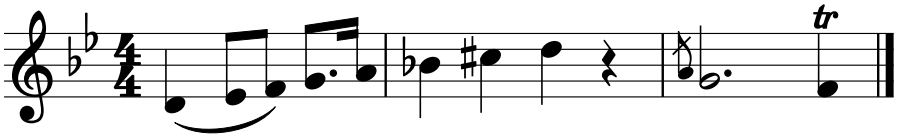
typst

```typst
#render-music(data, options: (font: "Petaluma"))
```

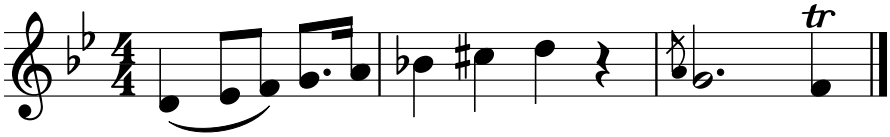
Petaluma



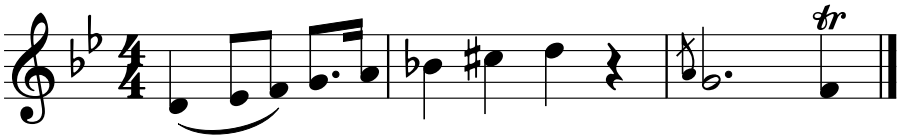
Leland



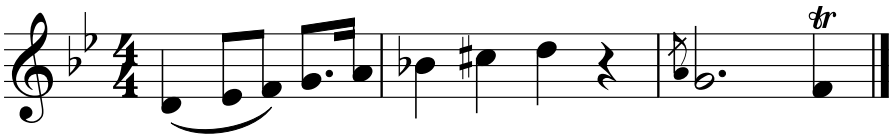
Gootville



Bravura



Leipzig



Supported Input Formats

Verovio auto-detects the input format for ABC, MusicXML, MEI, and Humdrum. For Volpiano and CMME, pass `inputFrom` explicitly.

All the files used in the examples are available in the project's Github.

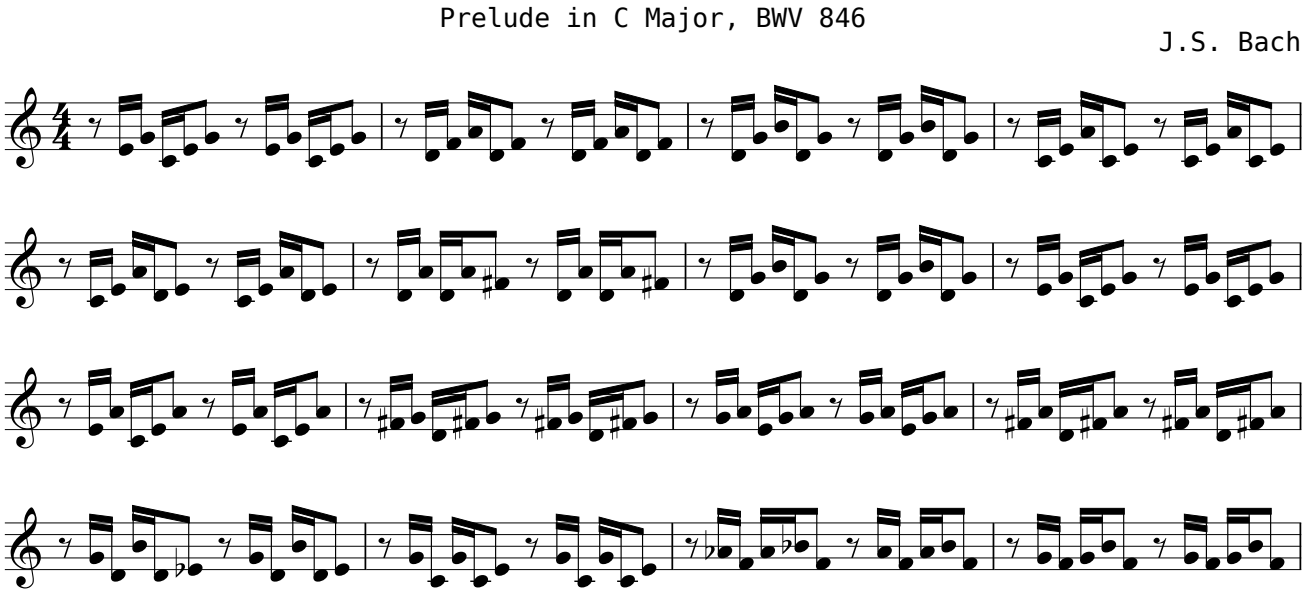
ABC Notation

ABC specification · Download ABC files

ABC is a compact text-based format popular for folk and classical melodies. Chord symbols are supported.

typst

```
#render-music(read("bach-prelude-cmaj.abc"))
```



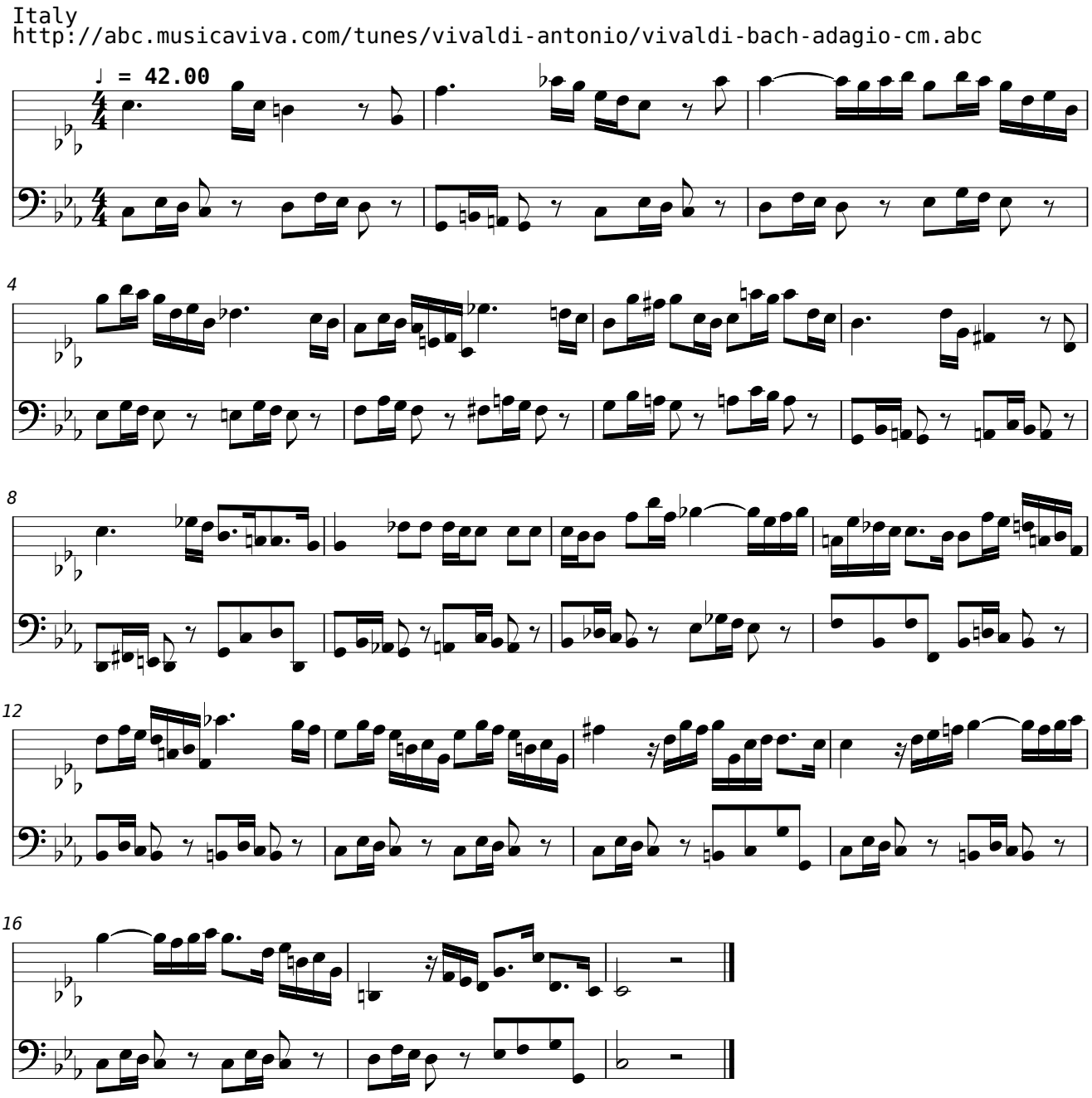
MusicXML

MusicXML 4.0 specification · Download MusicXML files

MusicXML is the standard interchange format for notation software. It supports grand staff, multiple voices, dynamics, and full score layout.

typst

```typst
#render-music(read("adagio.xml"))
```



MEI (Music Encoding Initiative)

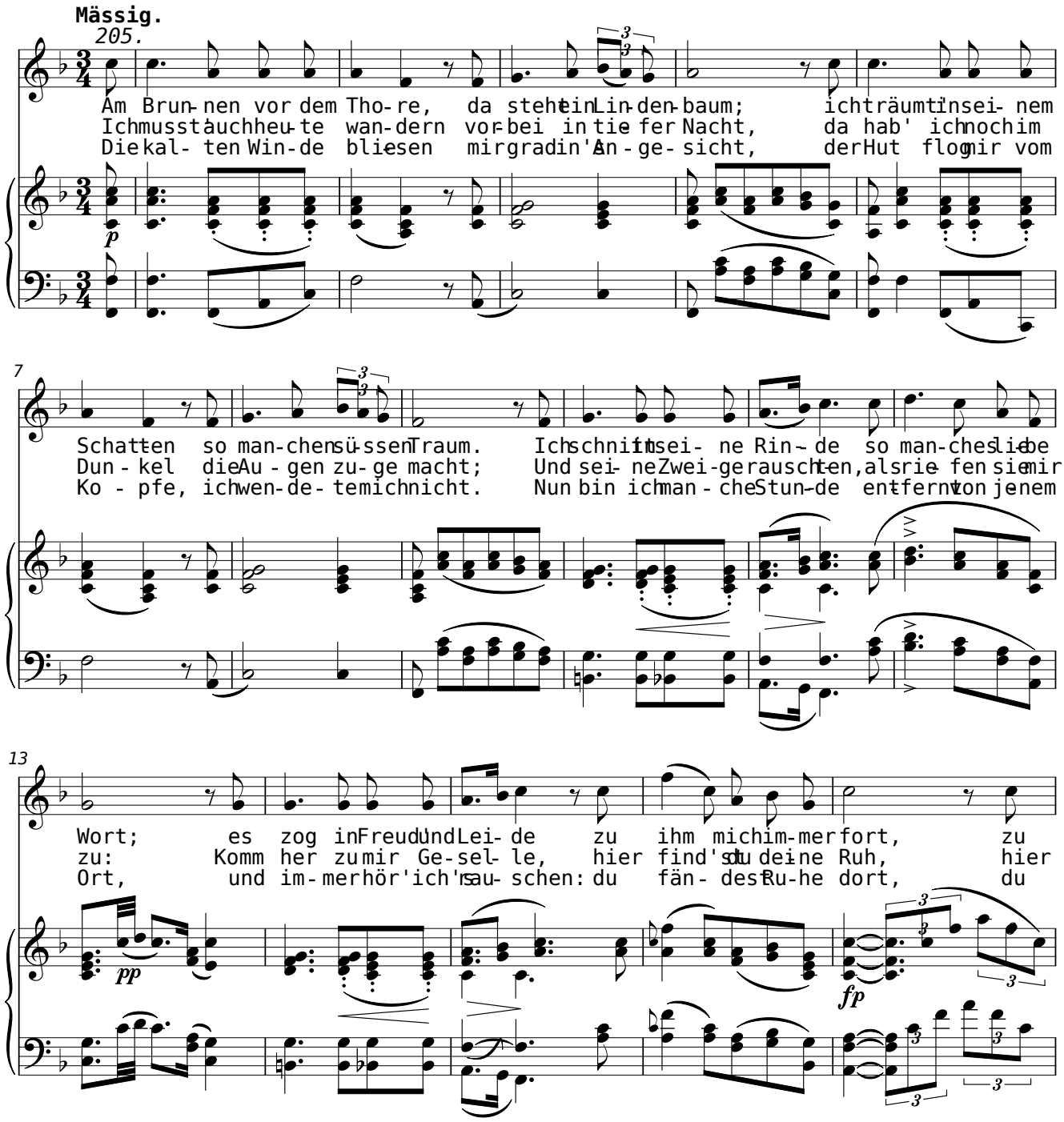
MEI Guidelines · Sample MEI encodings

MEI is a rich XML-based format used in musicology, supporting lyrics, polyphonic textures, fermatas, and detailed editorial markup.

typst

```
#render-music(read("schubert.mei"))
```

ndenbaum



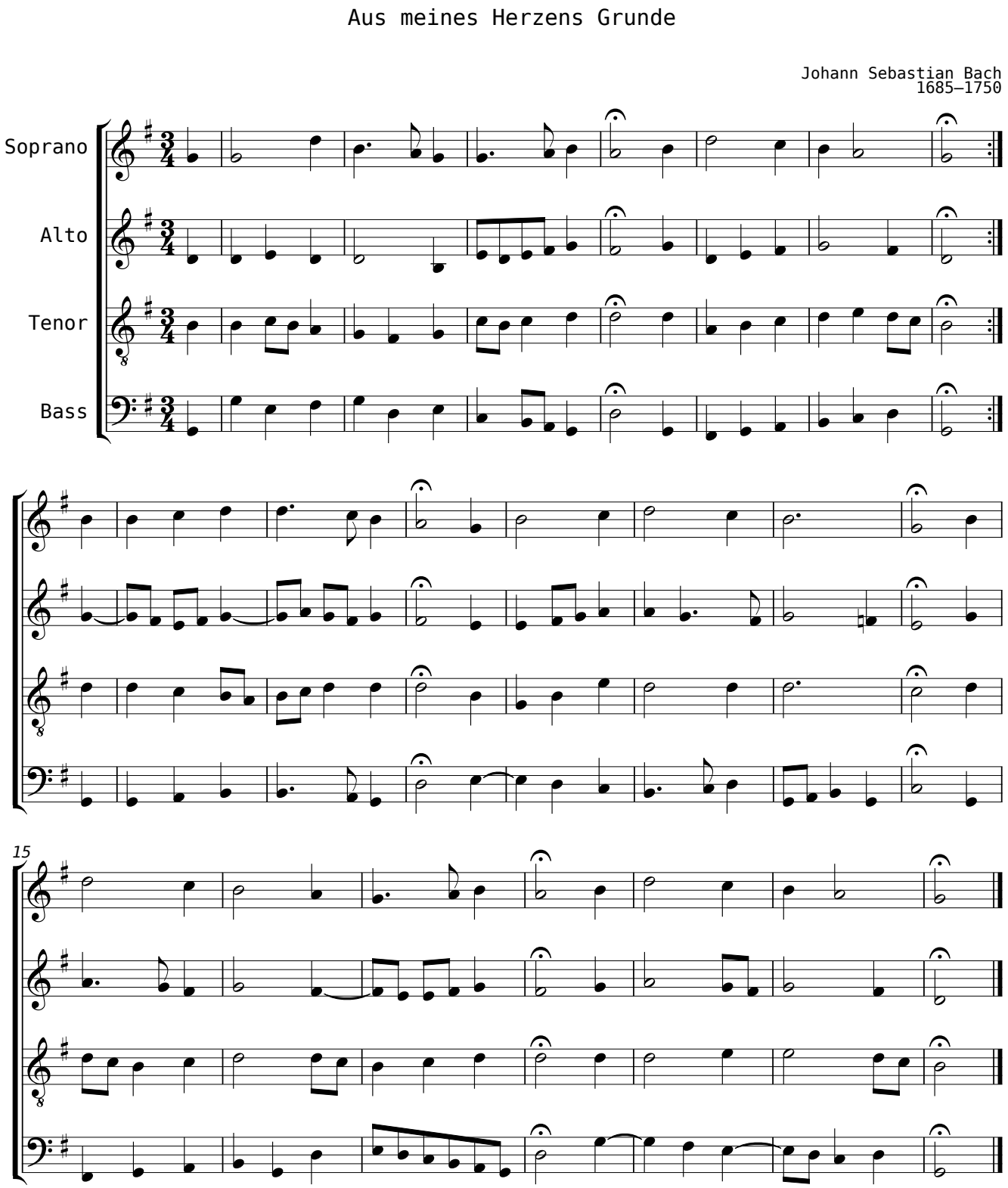
Humdrum

Humdrum User Guide · Download kern files

Humdrum uses a tab-separated spine structure with **kern encoding for pitches and durations. Widely used in computational musicology.

typst

```
#render-music(read("sample-humdrum.krn"))
```



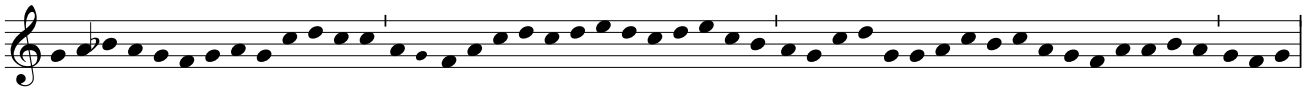
Volpiano

Volpiano specification · CANTUS database

Volpiano is a text encoding for medieval chant notation, used by the CANTUS database. Here is *Veni Creator Spiritus*, the famous Pentecost hymn. Requires `inputFrom: "volpiano"`.

typst

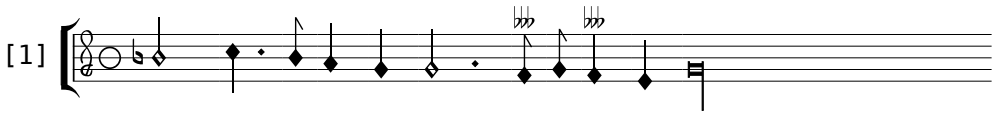
```
#render-music("1---g--hij---hgf--g--hg---k--lk--k7---hG--f---h--k--lk---l--m--l---k--lm---kj7--hg--kl---g--gh--k---
jk---h---gf--h--hjh7---g--f--g7---3", options: (inputFrom: "volpiano"))
```



CMME

CMME project · Download CMME files

CMME is an XML format for mensural notation (medieval and Renaissance music). Requires `inputFrom: "cmme"`.

typst

```
#render-music(read("cmme.xml"), options: (inputFrom: "cmme"))
```